

EMERGENCY DEPARTMENT / SIMULATOR

MED FLOW

Prioritize patients.
Optimize resources.

JUDGEABLE PROTOTYPE

Emergency-department resource management simulator

OPERATING VIEW / ARRIVALS • CAPACITY • DECISIONS

THE OPERATING CONSTRAINT

When demand spikes, the constraint is never just the queue

MEDFLOW makes the trade-offs visible before a real shift is under pressure.

SAME MINUTE / SAME OPERATING VIEW

CONSTRAINT	WHAT MUST BALANCE
01	Urgency and waiting time
02	Beds , staff, and transport
03	Policy against the live bottleneck

CONFIGURE A SCENARIO • SIMULATE THE SHIFT • OBSERVE THE BOTTLENECK • COMPARE THE POLICY.

One minute at a time, MEDFLOW turns arrivals into safe decisions

01

Generate

Arrivals follow a daily rhythm with an optional surge window.

02

Rank

Patients rise by live priority, not arrival order alone.

03

Dispatch

Ambulances and admissions require the full care set.

04

Release

Finished resources return to the operating pool.

05

Audit

Outcomes are measured and the run is checked.

ALL FIVE STAGES ARE VISIBLE IN THE DASHBOARD WORKFLOW.

Urgency leads. Waiting time prevents starvation.

THE SCORE RISES WHILE A PATIENT WAITS

$$\text{score}(t) = U[\text{triage}] + \alpha \cdot (t - \text{arrival}) + \gamma \cdot 60 / E[\text{stay}]$$

CLINICAL URGENCY + AGING TERM + OPTIONAL SHORT-STAY BONUS

SIGNAL

OPERATING MEANING

01

Five triage levels: resuscitation to non-urgent care.

02

Live score rises every minute waited.

03

Fixed-at-arrival heap key preserves exact ordering without re-sorting every minute.

Atomic allocation prevents half-treated patients and double-booked care

RESOURCE RULE	ALLOCATION BEHAVIOR	RELEASE POINT	SAFETY CHECK
CARE SET	Bed, nurse, and doctor together—or the patient waits.	When each resource is available.	No partial treatment.
POOL INTEGRITY	Reject over-capacity and duplicate allocations.	Resource-specific.	No double-booked care.
DOCTOR	Consult completes.	After consult.	Capacity returns.
BED + NURSE	Stay remains active.	After the stay.	Backfill usable capacity.
AMBULANCE	Handover completes.	After handover.	Blocked patients keep their place.

RESERVATION-BASED BACKFILLING KEEPS USABLE CAPACITY WORKING WITHOUT LETTING BLOCKED PATIENTS LOSE THEIR PLACE.

The dashboard shows the operating state, not just a final score

VIEW	WHAT IT SHOWS	WHY IT MATTERS
Live queue	Who is next, how long they have waited, and why they are blocked.	Make the next decision explainable.
Resource pressure	Current use, effective capacity, and peak utilization.	See the constraint forming.
Performance	Average wait, critical wait, target compliance, queue peak, throughput, and patients still waiting.	Keep outcomes visible.
Bottleneck detection	Turns a simulation into an operational conversation.	Connect the signal to the policy.

Stress tests turn “what if?” into a repeatable experiment

01

Emergency surge

Multiply arrivals during a defined window.

02

Staff shortage

Reduce doctor and nurse capacity without interrupting care already in progress.

03

Resource failure

Remove capacity from ICU beds, general beds, or ambulances.

04

Combined crisis

Stack demand and capacity shocks to expose the true bottleneck.

Same patients. Different policies. Honest trade-offs.

MEDFLOW replays the same patient stream across four policies, so differences come from the rule rather than luck.

NO.	POLICY	SAFETY	FAIRNESS	SPEED + UTILIZATION			
01	First come, first served	Arrival-led	Simple order	Baseline			
02	Strict triage	Urgency-led	Waiting risk	Priority speed			
03	Triage + aging the recommended default				Urgency first	Starvation control	Live trade-off
04	Triage + aging + bed turnover	Full care set	Capacity-aware	Utilization-aware			

THE COMPARISON VIEW MAKES SAFETY, FAIRNESS, SPEED, AND UTILIZATION VISIBLE TOGETHER.

The model is measurable, explainable, and independently checked

MEASURE	DEFINITION	INDEPENDENT ASSURANCE
Wait time	Queue minutes, including accrued wait for patients still waiting at shift end.	Rebuild from patient records.
Target limits	Explicit by triage: 2, 10, 30, 60, and 120 minutes.	Check each patient against the applicable limit.
Utilization	Mean in-use capacity divided by normal capacity.	Reconcile resource usage independently.
Audit	Every peak is checked against capacity.	Explainable run, independently checked.

REHEARSE THE SHIFT

DECISION SUPPORT, NOT A BLACK BOX

MEDFLOW gives hospital teams a safer way to rehearse the next shift

NO.	OPERATING MOVE
01	Start with a transparent simulation, not a black-box recommendation.
02	Show the constraint, compare the policy, and explain why a patient is waiting.
03	Next: connect real arrival patterns, calibrate service times, and turn scenario results into staffing and capacity decisions.

Prototype today. Decision support tomorrow.